Scarllette Ellis

PhD Researcher - Reinforcement Learning (she/her)

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A PhD student in **reinforcement learning**. I have experience in **research**, **public speaking**, **teaching** undergraduates and postgraduates, and **tutoring students**. My primary research is how autonomous agents can **learn high-level behaviour** and the **underlying mathematical structure** of the **agent-environment interaction**.

Education

Doctor of Philosophy in Computer Science Oct 2024 - Oct 2027 (expected)
University of Bath, UK
Supervisors: Professor Özgür Şimşek, Professor Guy McCusker

MRes in Accountable, Responsable, and Transparent AI Oct 2023-Oct 2024
University of Bath, UK
Supervisors: Professor Özgür Şimşek, Professor Guy McCusker
Dissertation: *Unifying Graph and Information Skill Discovery in Reinforcement Learning*

Masters in Mathematics and Computer Science Sep 2019 - Jul 2023
University of Bath, UK
Grade: 1st (Hons)
Masters Dissertation: *Getting Prepared: Finding Subgoals by using the Graphical Structure of Environments to Maximise Empowerment*
Supervisor: *Dr. Joshua B. Evans*
Undergraduate Dissertation: *Reinforcement Learning in the Board Game Pandemic*
Supervisor: *Dr. Joshua B. Evans*

Teaching and Supervision

Dissertation Supervision Sep 2024-Present
University of Bath, UK
Topics: Reinforcement learning, AI for tabletop games, Dynamic Difficulty Adjustment

- Marking of student work and providing feedback on their writing.
- Helping students solve theoretical and technical problems.
- Giving guidance to students on their project direction.

Tutor Sep 2022 - Present
University of Bath, UK
Teaching:

- Reinforcement Learning (Undergraduate & Masters Level)
- Bayesian Machine Learning (Masters Level)
- Artificial Intelligence (Undergraduate Level)

Responsibilities:

- Planning and delivering tutorials for undergraduate and postgraduate students.
- Marking coursework and providing feedback to students.
- Writing, running, and debugging automarking code.

Work Experience

Maths Support Coach - Future Foundation Maths Camps Sep 2026
Bootham School, York, UK

- Taught students aged 11-13 at the residential Maths Camp at Bootham Shool York.
- Ensured student safetey, acted as a role model for students, and showed them where studying maths can take you.

Lab Manager - Bath Reinforcement Learning Lab Sep 2025- Dec 2025
University of Bath, UK

- Organising and facilitating lab activities for more than 15 members.
- Activities include weekly lab meetings, paper discussions, and writing feedback sessions.

Automarking Internship Jun 2023–Sep 2023
University of Bath, UK

- Investigating the current state of automarking within the University of Bath Computer Science Department.
- Conducting interviews on how automarking was being used by lecturers.
- Delivering a seminar on how automarking was being used and providing advice on how best to use it, and the potential problems it has.

Lab Intern - Bath Reinforcement Learning Lab Jun 2022–Sep 2022
University of Bath, UK

- Investigating hierarchical reinforcement learning methods.
- Contributing to internal talks and paper discussions within the lab.

Head of TTRPGs - Tabletop Gaming Society Sep 2020-Jun 2021
University of Bath, UK

- Organising and running frequent society events.
- Running regular Dungeons & Dragons games.

Student Ambassador Sep 2020–Jun 2023
University of Bath, UK

- Providing university campus tours to prospective students.
- Giving advice on student life and insights on studying at university.

Skills

Conceptual: Reinforcement learning, Topology, linear algebra, machine learning
Technical: Python, Jax, NumPy, Matplotlib, LaTeX, Inkscape
Personal: Teaching, communication, delivering talks, supporting students

Talks and Events

Whose Science Is It Anyway? Using Improv theatre and Dungeons & Dragons to enhance science communication skills Jul 2025
BIG Event 2025 - Royal Institution London

Being Prepared: Creating Multi-Level Skill Hierarchies using Graph and Information Theory Oct 2024
University of Bath Centre for Mathematics and Algorithms for Data

The Current State and Best Practices for University Automarking Sep 2023
University of Bath Lecturer Training Day

Charity Dungeons and Dragons Live Stream Apr 2021
Off the Record Bath